EVIDENCE PACKET — NOTABLE CITATIONS

Citations to the applicant's publications

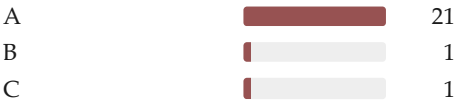
Akul Mehta — prepared for counsel in connection with the response to the Request for Evidence.

Generated 2026-11-07 from `notable-citations.sqlite` in this folder. 23 citations · 14 citing articles · 22 PDFs. Independence of each citing paper was established in the Independent Citation Record; this packet does not restate that analysis.

Summary

23 citations to the applicant's own publications, from 14 distinct citing articles, are documented in this packet. Each is placed at the section of the citing paper where the reference actually appears.

By tier



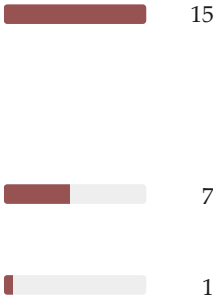
Tier A: the tool is the method performed, in Materials and Methods.

By applicant tool



One citing article may use more than one of the applicant's tools.

By research project



Project titles are quoted verbatim from the Research Background Inquiry Form. Which project a given paper belongs to was proposed by the preparer and is not taken from the Form — see the note at each project heading.

Screened out

38 candidate citations were examined and are *not* argued in this packet. They are listed in the appendix with the reason, so the screening is on the record:

- (unset) — 37
- not independent — 1

PDFs

23 of the 23 included citations have a PDF of the citing paper in this packet. Open items, including any citation still awaiting a PDF, are listed in `PACKET-STATUS.md`.

Index of citations

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
1	cit-000544	**Phylogenetic variations in a novel family of hyperstable apple snail egg proteins: insights into structural stability and functional trends** 2 CITATIONS	2019	29 Apr 2023 (bioRxiv preprint); 2024 (journal article, accepted 8 Jul 2024) earliest public: 29 Apr 2023 (preprint)	GLAD	Materials and Methods, p.4	Heras	article	preprint · webpage · article
	cit-000558	*as above*	2019	29 Apr 2023 (bioRxiv preprint); 2024 (journal article, accepted 8 Jul 2024) earliest public: 29 Apr 2023 (preprint)	GLAD	Materials and Methods, p.4	Heras	article	preprint · webpage · article
3	cit-000621	**LeGenD: High-throughput N-glycan profiling using explainable AI and lectin profiling**	2020	4 Jun 2026 earliest public: 30 Mar 2024 (preprint)	GlycoGlyph	Materials and Methods, p.12	Lewis	article	article
4	cit-000594	**AquIRE reveals the mechanisms of clinically induced RNA damage and the conservation and dynamics of glycoRNAs**	2020	2026 (accepted 18 Jan 2026) earliest public: 1 Apr 2025 (preprint)	GlycoGlyph	Materials and Methods, p.5	Knight	article	article

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
5	cit-001253	**Intelectin-2 is a broad-spectrum antimicrobial lectin**	2024	2026 (accepted 20 Nov 2025) earliest public: 26 Apr 2025	MAWE	Materials and Methods, p.15	Kiessling	article	article
6	cit-000518	**Major human schistosome species express different glycans with immunological and diagnostic implications** 2 CITATIONS	2019	2026 (accepted 3 Mar 2026) earliest public: 11 Jul 2025	GLAD	Materials and Methods, p.16	Hokke	article	article
	cit-000551	*as above*	2019	2026 (accepted 3 Mar 2026) earliest public: 11 Jul 2025	GLAD	Materials and Methods, p.16	Hokke	article	article
8	cit-000521	**Enzymatic tailoring of anionic glycans for characterizing lectin and antibody specificities in a microarray format**	2019	5 Nov 2025 earliest public: 25 Jul 2025	GLAD	Materials and Methods p.10; Results / Figures p.4	Wilson	article	article
9	cit-000629	**Glycan-reactive antibodies isolated from human HIV-1 vaccine trial participants show broad pathogen cross-reactivity** 2 CITATIONS	2020	10 Nov 2025 earliest public: 31 Jul 2025	GlycoGlyph	Materials and Methods p.20; Figure legend p.6	Georgiev	article	article

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
	cit-001261	*as above*	2024	10 Nov 2025 earliest public: 31 Jul 2025	MAWE	Materials and Methods p.20; Figure legend p.6	Georgiev	article	article
11	cit-000619	**Structural basis of secreted acid phosphatase polymerization in the Leishmania parasite**	2020	2026 (accepted 13 Apr 2026) earliest public: 23 Nov 2025	GlycoGlyph	Materials and Methods, p.14	Grossman-Haham	article	article
12	cit-000542	**Comparative phenotyping of surface markers and glycans in murine and human platelet-derived extracellular vesicles**	2019	2026 (accepted 13 Feb 2026) earliest public: 3 Dec 2025	GLAD	Materials and Methods p.5; Acknowledgments p.12	van der Wal	article	article
13	cit-000602	**Diminished sialoside binding in novel H5N1 influenza hemagglutinin variants identified in a human patient**	2020	2026 (accepted 1 Sep 2026) earliest public: 19 Jan 2026	GlycoGlyph	Materials and Methods, p.11	Subramaniam	article	article

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
14	cit-000516	**A Zebrafish Galectin-1 Isoform Is Expressed in Skin and Gills and Binds to Bacteria, Bacterial Adhesin Receptors, and Epidermal Mucus Glycans** 4 CITATIONS	2019	25 Apr 2026 earliest public: 31 Jan 2026	GLAD	Materials and Methods, p.21	Vasta	article	article
	cit-000603	*as above*	2020	25 Apr 2026 earliest public: 31 Jan 2026	GlycoGlyph	Materials and Methods, p.21	Vasta	article	article
	cit-000976	*as above*	2024	25 Apr 2026 earliest public: 31 Jan 2026	MAWE	Materials and Methods, p.21	Vasta	article	article
	cit-001252	*as above*	2024	25 Apr 2026 earliest public: 31 Jan 2026	MAWE	Materials and Methods, p.21	Vasta	article	article
18	cit-000613	**Transcriptome, glycome, and mucinome analysis reveal zinc is essential for the composition of mucus in the human goblet cell model HT-29-MTX**	2020	2026 (accepted 6 Aug 2026) earliest public: 30 Apr 2026	GlycoGlyph	Results / Figures p.4; Figure legend p.11	Maares	article	article

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
19	cit-001257	**Evaluating the role of non-HIV antigens in the initiation of QA013. 2 HIV bnAb lineage**	2024	2026 (accepted 5 Oct 2026; Article in Press) earliest public: 19 May 2026	MAWE	Materials and Methods, p.10	Overbaugh	article	article
20	cit-000416	**Glycan-mediated interactions of infectious hypodermal and hematopoietic necrosis virus (IHHNV) with host tissues in Pacific whiteleg shrimp (Penaeus vannamei)** 4 CITATIONS	2021	2026 (advance article; exact date not in the PDF) earliest public: Not shown in PDF	—	Discussion, p.19	Guérardel	article	article
	cit-000540	*as above*	2019	2026 (advance article; exact date not in the PDF) earliest public: Not shown in PDF	GLAD	Materials and Methods, p.30	Guérardel	article	article
	cit-000987	*as above*	2024	2026 (advance article; exact date not in the PDF) earliest public: Not shown in PDF	MAWE	Materials and Methods, p.29	Guérardel	article	article

#	ROW_UID	CITING PAPER	EARLIEST DATE OF CITED PAPER	CITING PAPER PUBLISHED	APPLICANT'S TOOL	SECTION & PAGE	SENIOR AUTHOR	ARTICLE TYPE	PDF IN PACKET
	cit-001259	*as above*	2024	2026 (advance article; exact date not in the PDF) earliest public: Not shown in PDF	MAWE	Materials and Methods, p.30	Guérardel	article	article

One numbered row is one citation. Where a single citing article cites more than one of the applicant's papers, it is numbered once and the additional citations are listed beneath it as "as above", each with its own `row_uid`.

A citation that appears in more than one place in the article lists every place, strongest first. A dagger (†) marks a row whose locations have not yet been re-read page by page, so only the single best-known location is shown.

The project titles below are quoted verbatim from the Research Background Inquiry Form. **The grouping of individual papers under those titles was proposed by the preparer** — the Form does not list which publication belongs to which project — and should be confirmed or corrected by the applicant before the packet is relied on.

Project 2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data

11 citing article(s), 17 citation(s).

Phylogenetic variations in a novel family of hyperstable apple snail egg proteins: insights into structural stability and functional trends

Maria Y Pasquevich, Marcos S Dreon, Maria E Diupotex-Chong, Horacio Heras
, 2024 · journals.biologists.com

SECTION I — BACKGROUND

Cited Paper(s)
GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays
2019

Additional citations from this citing paper
`cit-000558` — GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays (Materials and Methods, p.4)

Citing Paper
Phylogenetic variations in a novel family of hyperstable apple snail egg proteins: insights into structural stability and functional trends (article)
Earliest public date: 29 Apr 2023 (preprint) (preprint)

Project Number
2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data

Citing Authors' Institutions
Not recorded — confirm from the PDF's affiliation block.

Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 4 — Tier A — the tool is used in the Methods as the method performed · Compared

SECTION II — OVERVIEW OF USE

> *"The results of glycan specificity for PpaPV1 were compared with those of PsSC (Ituarte et al., 2018) and PdPV1 (Brola et al., 2020) using the Glycan Array Dashboard (glycotoolkit.com/GLAD/) (Mehta and Cummings, 2019)"*

Materials and Methods, p. 4 — Tier A — the tool is used in the Methods as the method performed

`cit-000558` — **additional citation of GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays**

> *"The results of glycan specificity for PpaPV1 were compared with those of PsSC (Ituarte et al., 2018) and PdPV1 (Brola et al., 2020) using the Glycan Array Dashboard (glycotoolkit.com/GLAD/) (Mehta and Cummings, 2019)"*

Materials and Methods, p. 4 — Tier A — the tool is used in the Methods as the method performed

Objectives

Pasquevich, Dreon, Diupotex-Chong and Heras investigate a novel family of hyperstable apple snail egg proteins (Perivitellin-1, PV1) from Pomacea snails, asking how protein stability relates to functional evolution. Their subject is PpaPV1 from the most basal lineage, Flagellata, which p.1 reports displays unparalleled structural and kinetic stability, ranking among the most hyperstable proteins documented in nature.

Importance to Field

The relationship between stability and functional evolution is, as the abstract on p.1 puts it, little explored for proteins purified from natural sources, so this is a rare natural dataset rather than an engineered one. p.10 develops the evolutionary argument that high stability may have provided tolerance to mutations, allowing new functions to be acquired during Pomacea diversification without compromising the native structure.

How and Why Your Work Was Used

The glycan-binding side of the study is where the applicant's tool enters. In Materials and Methods on p.4 the authors write that "The results of glycan specificity for PpaPV1 were compared with those of PsSC (Ituarte et al., 2018) and PdPV1 (Brola et al., 2020) using the Glycan Array Dashboard (glycotoolkit.com/GLAD/) (Mehta and Cummings, 2019)". The tool is named by full name, by URL, and by the applicant's author citation, and its role is comparison: it is how the lectin specificity of this basal protein was set against the previously published specificities of related proteins from other clades. That comparison is what supports the abstract's claim on p.1 that PpaPV1 has a milder lectin activity with narrower specificity than PV1s from the closely related Bridgesii clade. This folder also holds the earlier version of the work, which matters because the date of the statement relative to the applicant's petition date is what the extra files were collected to establish.

Findings / Conclusions / Recommendation

The paper concludes on p.1 that there is evidence for substantial structural and functional changes throughout the genus's PV1 evolution, with structural and kinetic stability decreasing in a clade-related fashion alongside large variation in defensive traits — the pale PpaPV1's lectin turning potent in the Bridgesii clade, adversely affecting gut morphology. p.10 frames this as stability enabling functional evolution.

FILES

- 01-preprint-Heras-2023-04-29.pdf PREPRINT — dated 2023-04-29 (Citation-summary.docx (earliest-date line))
- 02-webpage-Heras-2023.pdf WEBPAGE — dated 2023-04-29 (bioRxiv page printed in this PDF: "Posted April 29, 2023.")
- 03-article-Heras-2024.pdf ARTICLE — dated 2024-01-18 (Article page 1 printed in this PDF: "Received 18 January 2024; Accepted 8 July 2024" (the submission date, which is the earliest journal-side date; publication followed the 25 Apr 2024 filing))

`cit-000544` · additional rows cit-000558 · Scholar exhibit *glad-glycan-array-dashboard-a-visual-analytics-tool-for-glyc-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

LeGenD: High-throughput N-glycan profiling using explainable AI and lectin profiling

Haining Li, Angelo G Peralta, Sanne Schoffelen, Anders Holmgaard Hansen, Johnny Arnsdorf, Song-Min Schinn, Jonathan Skidmore, Biswa Choudhury, Frances Rocamora, Mousumi Paulchakrabarti
Journal of Biological Chemistry, 2026 · www.jbc.org

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	none
Citing Paper	LeGenD: High-throughput N-glycan profiling using explainable AI and lectin profiling (article) Earliest public date: 30 Mar 2024 (preprint) (preprint)
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 12 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"The glycan structures were drawn using GlycoGlyph (112)"*

Materials and Methods, p. 12 — Tier A — the tool is used in the Methods as the method performed

Objectives

Li, Peralta, Schoffelen, Lewis and colleagues present LeGenD, a method for high-throughput N-glycan profiling using explainable artificial intelligence combined with lectin profiling. The abstract on p.1 frames the objective as increasing the throughput and interpretability of glycan profiling, and the workflow figure on p.4 sets out the curated training set, the glycoform and glycoprotein outputs, and the lectin-derived features the model learns from.

Importance to Field

The authors make the field-level case themselves in their summary on p.8: "as awareness of the importance of glycosylation increases, new technologies are needed to increase the accessibility and throughput of glycan research tools". They also report on p.8 that SHAP analysis enhances understanding of the data beyond the raw weights of the neural network, which

addresses the interpretability objection that usually limits machine-learning methods in this area.

How and Why Your Work Was Used

The Materials and Methods on p.12 states that "The glycan structures were drawn using GlycoGlyph (112)", with the tool's URL given, reference 112 on p.15 being the applicant's GlycoGlyph paper. In a paper whose contribution is a computational pipeline for glycan profiling, the applicant's application is the named source of the glycan structure representations — the visual vocabulary in which the method's results are communicated. It is a Methods sentence in a paper building glycomics tooling, which is the most directly relevant kind of venue for this citation.

Findings / Conclusions / Recommendation

The paper reports that LeGenD performs high-throughput N-glycan profiling with explainable AI, and its summary on p.8 is that the explainability layer (SHAP) adds understanding beyond the network's weights. Its recommendation, also on p.8, is that the field needs more accessible and higher-throughput glycan tools — the argument the applicant's own tools were built to answer.

FILES

- 01-article-Lewis-2026.pdf ARTICLE

`cit-000621` · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

AquIRE reveals the mechanisms of clinically induced RNA damage and the conservation and dynamics of glycoRNAs

Zijian Zhang, Zornitsa Vasileva Kotopanova, Kexin Dang, Xiangxu Kong, Nicole Simms, Tin Wai Yuen, Lan Lam, Lauren Forbes Beadle, Emma Hilton, Taqdees Qureshi
Nucleic Acids Research, 2026 · academic.oup.com

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	none
Citing Paper	AquIRE reveals the mechanisms of clinically induced RNA damage and the conservation and dynamics of glycoRNAs (article) Earliest public date: 1 Apr 2025 (preprint) (preprint)
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 5 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"Glycans were drawn using GlycoGlyph [30]"*

Materials and Methods, p. 5 — Tier A — the tool is used in the Methods as the method performed

Objectives

Zhang, Kotopanova, Dang and colleagues publish in Nucleic Acids Research a Methods article introducing AquIRE, an assay for clinically induced RNA damage, which they then apply to the conservation and dynamics of glycoRNAs. The abstract on p.1 sets both halves: the damage-detection method, and what applying it reveals about glycoRNAs.

Importance to Field

GlycoRNAs are a contested and technically difficult subject, and the value of a new assay here is that it can detect species that earlier methods missed. The authors' own summary on p.12 is that "using the sensitivity of our AquIRE assay in conjugation with lectins allows us to detect glycoRNAs that display similar molecular biology to those described previously", which is a claim about both the assay and the biology.

How and Why Your Work Was Used

On p.5, in the Materials and Methods describing how the glycan-related figures were made, the authors state simply that "Glycans were drawn using GlycoGlyph [30]", reference 30 on p.17 being the applicant's GlycoGlyph paper. The sentence is short and unambiguous: the glycan structures that appear in this Nucleic Acids Research Methods paper were produced with the applicant's application rather than drawn by hand or with another package. GlycoGlyph's purpose is exactly this — drawing and naming glycan structures — so a Methods sentence assigning that task to it is the tool doing the job it was published to do.

Findings / Conclusions / Recommendation

The paper reports the mechanisms by which clinically induced RNA damage can be detected and applies them to glycoRNA conservation, concluding on p.12 that the lectin-combined assay detects glycoRNAs with molecular biology consistent with previously described ones.

FILES

- 01-article-Knight-2026.pdf ARTICLE

`cit-000594` · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Major human schistosome species express different glycans with immunological and diagnostic implications

Laudine MC Petralia, Angela van Diepen, Tao Zhang, Anna O Kildemoes, Tom Veldhuizen, D Linh Nguyen, Cornelis H Smit, Peter G Kremsner, Manfred Wuhrer, Jeremy M Foster
Nature Communications, 2026 · www.nature.com

SECTION I — BACKGROUND

Cited Paper(s)	GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays 2019

Additional citations from this citing paper	`cit-000551` — GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays (Materials and Methods, p.16)
Citing Paper	Major human schistosome species express different glycans with immunological and diagnostic implications (article) Earliest public date: 11 Jul 2025
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 16 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"Finally, GLycan Array Dashboard (GLAD) 93, version 3.1, from the glycotoolkit website (glycotoolkit.com/Tools/GLAD/) hosted by the National Center for Functional Glycomics, was used for data visualization, data mining, generation of heatmaps and boxplot graphs"*

Materials and Methods, p. 16 — Tier A — the tool is used in the Methods as the method performed

`cit-000551` — additional citation of GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays

> *"Finally, GLycan Array Dashboard (GLAD) 93, version 3.1, from the glycotoolkit website (glycotoolkit.com/Tools/GLAD/) hosted by the National Center for Functional Glycomics, was used for data visualization, data mining, generation of heatmaps and boxplot graphs"*

Materials and Methods, p. 16 — Tier A — the tool is used in the Methods as the method performed

Objectives

Petralia, van Diepen, Foster, Hokke and colleagues set out to characterise the glycan repertoire of Schistosoma haematobium, the species responsible for most human schistosome infection but, as p.1 puts it, largely unstudied compared with S. mansoni, and to ask whether the two species differ in ways that matter immunologically and diagnostically.

Importance to Field

The stated gap is clinically meaningful: S. haematobium causes the majority of infections and its own associated pathology, yet the in-depth glycome work had been done on a different species. p.1 reports that they found substantial differences from S. mansoni in core structure, fucosylation patterns and glucuronic acid modifications of glycosphingolipid glycans, which means the widely transferred assumptions from S. mansoni do not carry over.

How and Why Your Work Was Used

The article's Materials and Methods describes the glycan array work, and on p.16 the authors write that "Finally, GLycan Array Dashboard (GLAD) 93, version 3.1, from the glycotoolkit website (glycotoolkit.com/Tools/GLAD/) hosted by the National Center for Functional Glycomics, was used for data visualization, data mining, generation of heatmaps and boxplot graphs". The named tool is the applicant's, cited by reference 93 (the GLAD Bioinformatics paper, listed on p.19), and the sentence assigns it four specific analytical jobs — visualisation, data mining, and the two graph types in which the array results are presented.

Findings / Conclusions / Recommendation

The authors conclude on p.1 that S. haematobium and S. mansoni are differentially glycosylated and that S. haematobium glycans are involved in parasite-host immunobiology, reporting that IgG from S. haematobium-infected individuals preferentially binds the acidic GSL glycans relative to IgG from S. mansoni-infected individuals. They identify specific, immunogenic S. haematobium glycan targets, which is the diagnostic implication stated in the abstract.

FILES

- 01-article-Hokke-2026.pdf ARTICLE

`cit-000518` · additional rows cit-000551 · Scholar exhibit *glad-glycan-array-dashboard-a-visual-analytics-tool-for-glyc-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Enzymatic tailoring of anionic glycans for characterizing lectin and antibody specificities in a microarray format

Nuschy L, Wilson IBH
Journal of Biological Chemistry, 2025 · www.jbc.org

SECTION I — BACKGROUND

Cited Paper(s)	GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays 2019
Additional citations from this citing paper	none
Citing Paper	Enzymatic tailoring of anionic glycans for characterizing lectin and antibody specificities in a microarray format (article) Earliest public date: 25 Jul 2025
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 10 — Tier A — the tool is used in the Methods as the method performed · Implemented **Results / Figures**, page 4 — Tier B — the tool is used in the Results or Figures · Implemented

SECTION II — OVERVIEW OF USE

> *"the former generated using the GLAD tool (59) representing the relative fluorescent units (RFU) normalized to 70,000 as the maximum value"*

Materials and Methods, p. 10 — Tier A — the tool is used in the Methods as the method performed

> *"The heat map was generated using the GLAD tool and represents the relative fluorescent units (RFU) normalized to 70,000 as the maximum value"*

Results / Figures, p. 4 — Tier B — the tool is used in the Results or Figures

Objectives

Nuschy and Wilson, writing in the Journal of Biological Chemistry, set out to extend their own defined glycan microarrays to a library of natural anionic structures — N-acetylneuraminic acid and N-glycolylneuraminic acid, plus sulfated galactose, N-acetylglucosamine and glucuronic acid — built with recombinant glycosyl- and sulfo-transferases, and to compare anionic modifications on linear tetrasaccharides against typical biantennary N-glycan cores. p.2 states the goal: to analyse biantennary and linear glycans for accessibility and minimal requirements for lectin binding, and to re-examine anionic-epitope-recognising lectins and antibodies across different sialic acid types.

Importance to Field

The paper's opening on p.1 makes the case that this work is still needed: despite many studies of commercial lectin binding patterns, "unexpected recognition motifs are still revealed", and these reagents are used for rapid screening of glycosylation patterns and for diagnostic histochemistry. So a specificity that turns out narrower than assumed has consequences for anyone reading a diagnostic slide with that lectin.

How and Why Your Work Was Used

GLAD is named twice, in the two places a reader would look. In the figure captions on p.4 and repeated on p.5, the authors state that "The heat map was generated using the GLAD tool and represents the relative fluorescent units (RFU) normalized to 70,000 as the maximum value" — so the heat-map images that carry the binding results are output of the applicant's software. And in Materials and Methods, under "Microarray printing, analysis, and data processing" on p.10, they write that the data "are depicted as heat-maps (Figs. 3-5, summarized in Fig. S6) and graphs" and that "the former generated using the GLAD tool (59) representing the relative fluorescent units (RFU) normalized to 70,000 as the maximum value", reference 59 being the GLAD paper listed on p.12. This is the tool producing the reported figures, described in the methods section with a formal citation, and the same normalisation the authors chose is the one GLAD applies.

Findings / Conclusions / Recommendation

The authors report on p.6 that they "conclude that sulfate is not required for efficient antibody binding on a microarray, but rather the β4-linkage of the Gal to which the glucuronic acid is attached is important", and their summary on p.6 is that for most lectins and glycan-binding proteins there were no severe differences between biantennary N-glycans, Lewis-type glycans and linear saccharides.

FILES

- 01-article-Wilson-2025.pdf ARTICLE

`cit-000521` · Scholar exhibit *glad-glycan-array-dashboard-a-visual-analytics-tool-for-glyc-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Glycan-reactive antibodies isolated from human HIV-1 vaccine trial participants show broad pathogen cross-reactivity

Parker J. Jamieson, Xiaoying Shen, Alexandra A. Abu-Shmais, Perry T. Wasdin, Katarzyna Janowska, Robert J. Edwards, Garrett Scapellato, Maurice Bukenya, Lindsay E. Bass, Simone I. Richardson, Nelia P. Manamela, Shuying Liu, Maggie Barr, Lindsey Adams, Cristina Paola Velez-Castro, Caitlin McCarthy, Caroline A. Alexander, Rebecca A. Gillespie, Jessica

Mimms, Naveenchandra Suryadevara, Ty A. Sornberger, Seth J. Zost, Robert Parks, Shelby Flaherty, Alexis K. Janke, Bethany N. Howard, Yukthi P. Suresh, Ruth M. Ruprecht, James E. Crowe, Robert H. Carnahan, Justin R. Bailey, Masaru Kanekiyo, Daniel Lingwood, Barton F. Haynes, Penny L. Moore, Rachel H. Bonami, Georgia D. Tomaras, Priyamvada Archarya, David C. Montefiori, Spyros A. Kalams, Shan Lu, Ivelin S. Georgiev
, 2025 · journals.asm.org

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	`cit-001261` — Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process (Materials and Methods p.20; Figure legend p.6)
Citing Paper	Glycan-reactive antibodies isolated from human HIV-1 vaccine trial participants show broad pathogen cross-reactivity (article) Earliest public date: 31 Jul 2025
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 20 — Tier A — the tool is used in the Methods as the method performed · Implemented **Figure legend**, page 6 — Tier B — the tool is used in the Results or Figures · Implemented

SECTION II — OVERVIEW OF USE

> *"Figures of glycan structures were gathered using GlycoGlyph (49)"*

Materials and Methods, p. 20 — Tier A — the tool is used in the Methods as the method performed

> *"Diagrams of glycan structures were generated using GlycoGlyph (49)"*

Figure legend, p. 6 — Tier B — the tool is used in the Results or Figures

`cit-001261` — additional citation of Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process

> *"Antibodies were tested against a panel of 562 unique glycan structures following a previously reported microwave-assisted wet-erase protocol, but with several exceptions (50)"*

Materials and Methods, p. 20 — Tier A — the tool is used in the Methods as the method performed

> *"Glycan microarray protocol was followed using the microwave-assisted wet-erase process by Mehta et al. (50)"*

Figure legend, p. 6 — Tier B — the tool is used in the Results or Figures

Objectives

Jamieson, Shen, Abu-Shmais and Georgiev and colleagues isolate glycan-reactive antibodies from human HIV-1 vaccine trial participants and characterise their pathogen cross-reactivity. p.1 states the motivation: understanding the B-cell repertoire that vaccination can engage in humans is crucial for developing preventive vaccines, and the study works from peripheral blood mononuclear cells of trial participants.

Importance to Field

The glycan shield of HIV-1 has been a long-standing obstacle to antibody-based vaccine design. The authors' proposal, on p.3, inverts it: "we propose that the elicitation of glycan-reactive antibodies can serve as an effective strategy for turning the elusive HIV-1 glycan shield into a Trojan horse", and they find on p.7 that although the antibodies are autoreactive, their overall reactivity toward self-antigen is limited, which is what makes the class safe to pursue. That safety finding is a prerequisite for the vaccine strategy, not an afterthought.

How and Why Your Work Was Used

Both of the applicant's contributions appear in this paper's array work, in the Methods and in a figure legend. In the Materials and Methods on p.20, under "Glycan microarray", the authors write that "Antibodies were tested against a panel of 562 unique glycan structures following a previously reported microwave-assisted wet-erase protocol, but with several exceptions (50)" — reference 50 being the applicant's MAWE paper — and the same paragraph ends "Figures of glycan structures were gathered using GlycoGlyph (49)". The figure legend on p.6 repeats both: "Diagrams of glycan structures were generated using GlycoGlyph (49). Glycan microarray protocol was followed using the microwave-assisted wet-erase process by Mehta et al. (50)." So the assay that generated the paper's central binding data followed the applicant's published MAWE protocol, with the authors stating their deviations honestly, and the glycan diagrams in the figures came from the applicant's application.

Findings / Conclusions / Recommendation

The authors report that glycan-reactive antibodies from vaccine trial participants show broad pathogen cross-reactivity, and on p.3 that such a public class of antiviral antibody recognition can be used as a target for broadly reactive vaccine candidates against HIV-1. On p.7 they conclude that the HVTN124 glycan-reactive monoclonal antibodies are favourable and safe for further elicitation in future HIV-1 vaccine clinical trials.

FILES

- 01-article-Georgiev-2025.pdf ARTICLE

`cit-000629` · additional rows cit-001261 · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Structural basis of secreted acid phosphatase polymerization in the Leishmania parasite

Priyanka Bose, Irit Dahan, Alexander Upcher, Ran Zalk, Iris Grossman-Haham
Protein Science, 2026 · onlinelibrary.wiley.com

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	none
Citing Paper	Structural basis of secreted acid phosphatase polymerization in the Leishmania parasite (article) Earliest public date: 23 Nov 2025

Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 14 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"A model of the N-linked glycan containing two N-acetylglucosamine (GlcNac) units and a mannose (Man) was generated in GlycoGlyph (Mehta & Cummings, 2020)"*

Materials and Methods, p. 14 — Tier A — the tool is used in the Methods as the method performed

Objectives

Bose, Dahan, Upcher, Zalk and Grossman-Haham determine the structural basis of polymerisation of the secreted acid phosphatase (SAP) of Leishmania, an abundant extracellular glycoprotein and major virulence factor in leishmaniasis. p.1 sets up the puzzle: enzymes that form filaments usually switch between proto-meric and polymeric states with cellular conditions, whereas SAP either forms highly stable extracellular filaments or stays globular, with no reversible interconversion.

Importance to Field

Leishmaniasis is a neglected tropical disease and SAP is regarded as a major virulence factor, so the assembly mechanism is not merely a structural curiosity. The binary assembly pattern described on p.1 is unusual enough that explaining it bears on how the parasite's extracellular machinery works, and the paper's interface analysis on p.11 accounts for the stability of the filaments across diverse chemical conditions.

How and Why Your Work Was Used

The Materials and Methods on p.14 records the use: "A model of the N-linked glycan containing two N-acetylglucosamine (GlcNac) units and a mannose (Man) was generated in GlycoGlyph (Mehta & Cummings, 2020)". The applicant's application is named as the tool that produced a specific structural model — a defined N-linked glycan with two GlcNAc units and a mannose — cited directly to the applicant's paper in the body text rather than only by number. This is the tool used to render a molecular object that appears in the paper's structural account of a glycoprotein, which is the natural use-case for a glycan drawing and naming application.

Findings / Conclusions / Recommendation

The authors conclude on p.10 that "SAP polymerization is governed by a combination of sequence-specific interactions within the β-hairpin motif rather than solely by the presence of the motif", and on p.11 that the interactions they identify create an extensive interface accounting for the high stability of LmSAP filaments across diverse chemical conditions. They also report on p.7 that glycosylation at the site they examined is not crucial for SAP assembly, solubility or function.

FILES

- 01-article-GrossmanHaham-2026.pdf ARTICLE

`cit-000619` · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Comparative phenotyping of surface markers and glycans in murine and human platelet-derived extracellular vesicles

Olga An, Friedrich Reusswig, Viola Krenzlin, Carsten Deppermann, Dianne E van der Wal
, 2026 · www.sciencedirect.com

SECTION I — BACKGROUND

Cited Paper(s)	GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays 2019
Additional citations from this citing paper	none
Citing Paper	Comparative phenotyping of surface markers and glycans in murine and human platelet-derived extracellular vesicles (article) Earliest public date: 3 Dec 2025
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 5 — Tier A — the tool is used in the Methods as the method performed · Implemented **Acknowledgments**, page 12 — Tier D — acknowledged

SECTION II — OVERVIEW OF USE

> *"Glycan Array Dashboard (version 3.1) was used to analyze and visualize the glycan array data [44]"*

Materials and Methods, p. 5 — Tier A — the tool is used in the Methods as the method performed

> *"We also thank Akul Mehta (Harvard University) for valuable discussions and help with the glycan tool Glycan Array Dashboard"*

Acknowledgments, p. 12 — Tier D — acknowledged

Objectives

An, Reusswig, Krenzlin, Deppermann and van der Wal set out to compare murine and human platelet-derived extracellular vesicles (PEVs) by surface markers and glycans, because mouse models are used to study PEVs while the two species' PEV glycan surfaces had not been systematically compared. The abstract on p.1 frames PEVs as important carriers of signalling molecules, lipids and proteins, and the study's glycan arm uses a glycan array alongside flow cytometry.

Importance to Field

The translational point is that conclusions drawn in mouse do not automatically transfer to human. p.3 reports that the findings "highlight the heterogeneity of PEV populations that show distinct interactions with immune cells", and the discussion on p.12 works through the consequences — lower surface β-galactose and fucose exposure on murine vesicles, higher surface mannose on human ones, and what each would mean for interaction with macrophage galactose lectin or the mannose receptor CD206.

How and Why Your Work Was Used

The article's use of the applicant's work is recorded twice, in two different registers. Under the Methods heading "2.6 | Glycan array analysis" on p.5 the authors write that "Glycan Array Dashboard (version 3.1) was used to analyze and visualize the glycan array data [44]", reference 44 on p.14 being "Mehta AY, Cummings RD. GLAD: GLycan Array Dashboard..." — a Methods use, versioned, cited to the applicant's paper, and assigned the analysis and visualisation of the array data. Separately, the Acknowledgments on p.12 record that "We also thank Akul Mehta (Harvard University) for valuable discussions and help with the glycan tool Glycan Array Dashboard and Consortium for Functional Glycomics (CFG) names". That second sentence is a courtesy and is not counted as a use of the tool; it is included because it shows the authors went to the applicant for help with the tool by name.

Findings / Conclusions / Recommendation

The authors report species differences in PEV surface glycans and markers, and on p.11 they note that array and flow-cytometry observations differ, which they attribute to the microarray detecting total rather than surface glycan content — an honest limitation stated in the paper's own words. Their broader point, on p.12, is that these glycan differences plausibly change how PEVs are recognised by hepatocytes, fibroblasts and neurons.

FILES

- 01-article-VanDerWal-2026.pdf ARTICLE

`cit-000542` · Scholar exhibit *glad-glycan-array-dashboard-a-visual-analytics-tool-for-glyc-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Diminished sialoside binding in novel H5N1 influenza hemagglutinin variants identified in a human patient

John H Ni, Saeid Malek Zadeh, Alison M Berezuk, Ryan Lynam, Peter Axerio-Cilies, Xing Zhu, Katharine S Tuttle, Gethin Rh Owen, Maria Tokuyama, Sriram Subramaniam
Nature Communications, 2026 · www.nature.com

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	none
Citing Paper	Diminished sialoside binding in novel H5N1 influenza hemagglutinin variants identified in a human patient (article) Earliest public date: 19 Jan 2026
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 11 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"We produced glycan schematics using GlycoGlyph then adjusted them on Inkscape"*

Materials and Methods, p. 11 — Tier A — the tool is used in the Methods as the method performed

Objectives

Ni, Malek Zadeh, Berezuk, Subramaniam and colleagues study the haemagglutinin of the H5N1 influenza strain that hospitalised an adolescent patient in British Columbia in 2024 (BC24). p.1 states what they show: that the N169 α2,3-linked auto-glycan found in the sialic acid binding site of previously studied H5 haemagglutinin proteins is absent in purified BC24 HA, suggesting greatly reduced affinity for α2,3-linked sialosides.

Importance to Field

This is a human case of a novel zoonotic influenza strain, so the receptor-binding question is directly about pandemic potential — whether the virus can engage human-type receptors. The paper addresses that on pp.8-9, reporting that the reduced-binding minority variants of interest remain functional and that weakly binding variants at least remained relatively fit, and noting on p.10 that concerns about near-term health emergencies from related strains are largely assuaged by retained antiviral susceptibility and a conserved stem region.

How and Why Your Work Was Used

GlycoGlyph appears in the Materials and Methods on p.11, where the authors describe producing the glycan figures: the structures in Figs. 3 and 4 were "produced on GlycoGlyph with adjustments on Inkscape", and the same section states "We produced glycan schematics using GlycoGlyph then adjusted them on Inkscape". The applicant's application is named as the origin of the glycan schematics in a methods paragraph, with the downstream cosmetic adjustment in a general vector editor noted honestly. The glycan microarray analyses that the paper reports on p.1 — showing severely reduced or absent binding to both α2,3- and α2,6-linked sialosides — are the data whose structures are drawn this way.

Findings / Conclusions / Recommendation

The authors conclude that the BC24 HA variants display severely reduced or no binding to both α2,3- and α2,6-linked sialosides, that full-length BC24 HA expressed in A549 alveolar carcinoma cells still drives membrane fusion albeit at lower levels than previous H5 HA proteins, and that post-infection patient sera bind BC24 HA strongly (p.1). Because each of the two mutations of interest independently and jointly reduces sialoside binding, they infer multiple populations of virus in this case.

FILES

- 01-article-Subramaniam-2026.pdf ARTICLE

`cit-000602` · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

A Zebrafish Galectin-1 Isoform Is Expressed in Skin and Gills and Binds to Bacteria, Bacterial Adhesin Receptors, and Epidermal Mucus Glycans

Chiguang Feng, Kelsey Abernathy, Sheng Wang, Guanghui Zong, Nilli Zmora, Allison Shupp, Muddassar Iqbal, Lai-Xi Wang, Gerardo R Vasta

International Journal of Molecular Sciences, 2026 · www.mdpi.com

SECTION I — BACKGROUND

Cited Paper(s)	GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays 2019
Additional citations from this citing paper	`cit-000603` — GlycoGlyph: a glycan visualizing, drawing and naming application (Materials and Methods, p.21) `cit-000976` — Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies (Materials and Methods, p.21) `cit-001252` — Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process (Materials and Methods, p.21)
Citing Paper	A Zebrafish Galectin-1 Isoform Is Expressed in Skin and Gills and Binds to Bacteria, Bacterial Adhesin Receptors, and Epidermal Mucus Glycans (article) Earliest public date: 31 Jan 2026
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 21 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"The results were analyzed by the GLycan Array Dashboard (GLAD) platform [128], and the glycan models were constructed using GlycoGlyph [129]"*

Materials and Methods, p. 21 — Tier A — the tool is used in the Methods as the method performed

`cit-000603` — additional citation of GlycoGlyph: a glycan visualizing, drawing and naming application

> *"The results were analyzed by the GLycan Array Dashboard (GLAD) platform [128], and the glycan models were constructed using GlycoGlyph [129]"*

Materials and Methods, p. 21 — Tier A — the tool is used in the Methods as the method performed

`cit-000976` — additional citation of Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies

> *"Glycan microarray analysis following established procedures [125-127] was carried out at the National Center for Functional Glycomics (NCFG) at Beth Israel Deaconess Medical Center, Harvard Medical School"*

Materials and Methods, p. 21 — Tier A — the tool is used in the Methods as the method performed

`cit-001252` — **additional citation of Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process**

> *"Glycan microarray analysis following established procedures [125-127] was carried out at the National Center for Functional Glycomics (NCFG) at Beth Israel Deaconess Medical Center, Harvard Medical School"*

Materials and Methods, p. 21 — Tier A — the tool is used in the Methods as the method performed

Objectives

Feng, Abernathy, Vasta and colleagues characterise a galectin-1 isoform from zebrafish (Drgal1-L2): where it is expressed, what it binds, and whether a galectin present in epidermal mucus can influence bacterial attachment. The abstract on p.1 states the scope — expression in skin and gills, and binding to bacteria, bacterial adhesin receptors and epidermal mucus glycans — and p.14 records that they first carried out the molecular and biochemical characterisation, then the tissue-specific expression, then the ligand identification, "with the aim of addressing the question" of the mucus galectin's role.

Importance to Field

The question sits at the boundary of innate immunity and glycobiology in a model organism whose mucosal immunity is less well mapped than the mouse's. On p.13 the authors confront the practical ambiguity directly, weighing whether bacteria bind mucus glycans through galactosyl-binding adhesins or whether endogenous galectins in the immobilised mucus bridge bacteria to each other, and conclude on p.13 that their model "b" is the one most likely to illustrate what happens in the environmental setting.

How and Why Your Work Was Used

Glycan microarray analysis is described in Materials and Methods §4.9.5 on p.21, which opens "Glycan microarray analysis following established procedures [125-127]" and names the National Center for Functional Glycomics at Beth Israel Deaconess Medical Center, Harvard Medical School, as the site where version 5.0 of the CFG Glycan Microarray was run. Reference 127 of that range is the applicant's MAWE paper (Mehta, Tilton, Muerner, von Gunten, Heimburg-Molinaro and Cummings, Glycobiology 2024), listed in full on p.31, so the applicant's reusable-array procedure is one of the three established procedures the analysis followed. The same §4.9.5 on p.21 closes with "The results were analyzed by the GLycan Array Dashboard (GLAD) platform [128], and the glycan models were constructed using GlycoGlyph [129]", and references 128 and 129 on p.31 are the applicant's GLAD and GlycoGlyph papers. The binding data shown in Fig. 5 (p.6, with the figure panel on p.7) were therefore produced on the applicant's array methodology, reduced in GLAD, and drawn in GlycoGlyph — three of the applicant's contributions in one Methods subsection.

Findings / Conclusions / Recommendation

The paper reports that Drgal1-L2 is expressed in skin and gills, binds bacteria and bacterial adhesin receptors, and recognises LacNAc-type galactosides, with the microarray confirming the preference on p.6. Its conclusion on p.13 is that galectin-mediated bridging in mucus is the more likely explanation of the bacterial-aggregation observations.

FILES

- 01-article-Vasta-2026.pdf ARTICLE

`cit-000516` · additional rows cit-000603;cit-000976;cit-001252 · Scholar exhibit *glad-glycan-array-dashboard-a-visual-analytics-tool-for-glyc-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Transcriptome, glycome, and mucinome analysis reveal zinc is essential for the composition of mucus in the human goblet cell model HT-29-MTX

Christoph Sch"ussler, Nara Chung, Renaud Leonard, Heike Sprenger, Tari R"odel, Keira E Mahoney, Susanne Thomsen, Esther Ocket, Claudia Matthaeus, Jordan Denis
, 2026 · www.nature.com

SECTION I — BACKGROUND

Cited Paper(s)	GlycoGlyph: a glycan visualizing, drawing and naming application 2020
Additional citations from this citing paper	none
Citing Paper	Transcriptome, glycome, and mucinome analysis reveal zinc is essential for the composition of mucus in the human goblet cell model HT-29-MTX (article) Earliest public date: 30 Apr 2026
Project Number	2 — Developing Bioinformatics Tools to Capture and Analyze Glycomics and Glycan Microarray Data
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Results / Figures**, page 4 — Tier B — the tool is used in the Results or Figures · Implemented **Figure legend**, page 11 — Tier B — the tool is used in the Results or Figures · Implemented

SECTION II — OVERVIEW OF USE

> *"Glycan structures were drawn with GlycoGlyph14"*

Results / Figures, p. 4 — Tier B — the tool is used in the Results or Figures

> *"Predicted N-glycan structures are shown above the corresponding m/z values. Glycans were drawn with GlycoGlyph14"*

Figure legend, p. 11 — Tier B — the tool is used in the Results or Figures

Objectives

Schüßler, Chung, Léonard, Maares and colleagues analyse transcriptome, glycome and mucinome data from the human goblet cell model HT-29-MTX to determine how zinc deficiency affects mucin synthesis and glycosylation. p.1 frames the problem: zinc deficiency affects roughly a billion people worldwide, compromises the intestinal defence barrier, and alters mucus, but the processes and extent of the effect on goblet cell mucin synthesis remain poorly understood.

Importance to Field

The stated gap on p.1 is that the mechanism behind zinc-deficiency-related loss of the mucus barrier is not understood, which matters because the clinical consequences — increased intestinal infections, inflammation, diarrhoea — are widespread. The paper's result, summarised on p.7, gives the field a specific molecular account: altered glycosylation machinery and increased MUC2 and MUC17 rather than a generic secretory failure.

How and Why Your Work Was Used

GlycoGlyph is named in the figures and their legends. The model of O-glycan biosynthesis in Fig. 1 carries the sentence on p.4, "Glycan structures were drawn with GlycoGlyph14", reference 14 being the applicant's GlycoGlyph paper. The same statement is repeated in a figure legend on p.11: "Predicted N-glycan structures are shown above the corresponding m/z values. Glycans were drawn with GlycoGlyph14". So the glycan structures that annotate the mass-spectrometry results in this multi-omics paper are drawn with the applicant's application, in both the schematic overview and the analytical glycan figure. The mentions sit in figure content rather than in the Materials and Methods text.

Findings / Conclusions / Recommendation

The authors report that zinc deprivation altered the goblet cell transcriptome, affecting genes involved in zinc transport, mucin synthesis and glycosylation, and that mucus composition changed, with significantly increased MUC2 and MUC17 at mRNA and protein level. Their summary on p.7 is that no differences in MUC5AC and MUC16 production appeared across conditions, but both the C-terminal fragment and mature MUC2 were significantly overexpressed in zinc-deficient cells and more MUC2 was secreted, confirming the mass-spectrometry result.

FILES

- 01-article-Maares-2026.pdf ARTICLE

`cit-000613` · Scholar exhibit *glycoglyph-a-glycan-visualizing-drawing-and-naming-applicati-citing-2026-11-01.pdf* (Independent Citation Record, not enclosed here)

Project 3 — Advancing Glycan Research with Microarray Innovations

2 citing article(s), 2 citation(s).

Intelectin-2 is a broad-spectrum antimicrobial lectin

Amanda E Dugan, Deepsing Syangtan, Eric B Nonnecke, Rajeev S Chorghade, Amanda L Peiffer, Jenny J Yao, Jessica Ille-Bunn, Dallis Sergio, Gleb Pishchany, Catherine Dhennezel
, 2026 · www.nature.com

SECTION I — BACKGROUND

Cited Paper(s)	Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process 2024
Additional citations from this citing paper	none
Citing Paper	Intelectin-2 is a broad-spectrum antimicrobial lectin (article) Earliest public date: 26 Apr 2025
Project Number	3 — Advancing Glycan Research with Microarray Innovations
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 15 — Tier A — the tool is used in the Methods as the method performed

SECTION II — OVERVIEW OF USE

> *"the core facility also changed protocols and began regenerating and reusing slides; the long-term effects on fluorescence intensity are unclear"*

Materials and Methods, p. 15 — Tier A — the tool is used in the Methods as the method performed

Objectives

Dugan, Syangtan, Nonnecke, Bevins and Kiessling investigate what intelectin-2 does. p.1 states that mammals regulate the composition and activity of their native microbiota at colonisation sites, that lectins at these sites influence microbial populations, but that their functional roles are often unclear; the study finds that mouse and human intelectin-2 engage and crosslink mucins through carbohydrate recognition, recognise microbes within native microbial communities from the respiratory and gastrointestinal tracts, and reduce the viability of the microbes they bind.

Importance to Field

Intelectins are found in chordates at mucosal barriers and their functions were poorly characterised, so establishing that intelectin-2 is a broad-spectrum antimicrobial lectin is a substantive change to the picture of innate mucosal defence. The authors' conclusions on p.6 are stated at that level: "Together, these data demonstrate that mItln2 recognizes specific microbes that colonize tissues in which mItln2 is expressed" and that mItln2 compromises the cellular integrity of bound bacteria and may influence cell viability.

How and Why Your Work Was Used

This row needs to be read with care. The microbial glycan array work is described in Materials and Methods on p.15, where the authors detail how arrays were probed and scanned and how replicate values were reduced; the reference to the applicant's MAWE paper (reference 69, listed in full on p.19 as "Mehta, A. Y. et al. Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process. Glycobiology 34, cwad091 (2024)") appears in the sentence: "in 2024, the core facility also changed protocols and began regenerating and reusing slides; the long-term effects on fluorescence intensity are unclear". That is a hedge attached to the practice of reusing slides, in a passage explaining why fluorescence intensities differ between experiments. It is a Methods-adjacent reference to the reusable-array work, not a statement that the MAWE protocol was followed as the method, so no use type is claimed for this row.

Findings / Conclusions / Recommendation

The paper concludes that intelectin-2 is a broad-spectrum antimicrobial lectin whose calcium-coordinated carbohydrate binding engages mucus and microbial surfaces, that bound microbes suffer loss of viability, and on p.7 that terminal amine-containing substituents may serve as an additional microbial ligand for the human protein.

FILES

- 01-article-Kiessling-2026.pdf ARTICLE

`cit-001253` · Scholar exhibit *reusable-glycan-microarrays-using-a-microwave-assisted-wet-e-citing-2026-10-30.pdf* (Independent Citation Record, not enclosed here)

Evaluating the role of non-HIV antigens in the initiation of QA013. 2 HIV bnAb lineage

Delphine M Depierreux, Kevin Sung, Lindsay A Bergan, Mackenzie Shipley, Yuzheng Zhang, Frederick A Matsen IV, Paul D Lampe, Julie Overbaugh
Scientific Reports, 2026 · www.nature.com

SECTION I — BACKGROUND

Cited Paper(s)	Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process 2024
Additional citations from this citing paper	none
Citing Paper	Evaluating the role of non-HIV antigens in the initiation of QA013. 2 HIV bnAb lineage (article) Earliest public date: 19 May 2026
Project Number	3 — Advancing Glycan Research with Microarray Innovations
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Materials and Methods**, page 10 — Tier A — the tool is used in the Methods as the method performed · Implemented

SECTION II — OVERVIEW OF USE

> *"Data were analyzed using the MAWE method as described previously 45"*

Materials and Methods, p. 10 — Tier A — the tool is used in the Methods as the method performed

Objectives

Depierreux, Sung, Bergan, Matsen, Lampe and Overbaugh test the hypothesis that maturation of the broadly neutralising antibody lineage QA013.2 may have been initiated by non-HIV antigens. p.1 explains why: breadth is usually achieved through affinity maturation, but the inferred naïve precursors of some bnAbs fail to bind the epitopes their mature counterparts target, so the initiating antigens remain elusive, which complicates vaccine design.

Importance to Field

The initiating-antigen problem is one of the central obstacles in HIV vaccine design, and the paper's approach — looking outside HIV for the priming exposure — reframes it. Their conclusion on p.2 is stated at that level of consequence: "Together, these findings suggest that eliciting optimal HIV antibodies may require immunogens beyond HIV-derived antigens alone."

How and Why Your Work Was Used

The glycan array work in this study is analysed with the applicant's method. In Materials and Methods on p.10 the authors state that "Data were analyzed using the MAWE method as described previously 45", reference 45 on p.14 being "Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process". The sentence is direct: the analysis followed the applicant's published MAWE method, cited to it. Where other citing articles name the applicant's work in a hedge or a figure legend, this one assigns the analysis of the array data to the applicant's method in a Methods sentence.

Findings / Conclusions / Recommendation

The authors report that the QA013.2 lineage acquired somatic mutations before HIV seroconversion and that if early maturation is indeed driven by non-HIV exposure, the required antigen may involve low-affinity interactions or complex conformational features or glycosylation patterns that remain challenging to detect in standard high-throughput screens (p.1). They also report on p.6 that some shared hits across screens do not represent a coherent motif-driven specificity, which is a candid limit on what their screens can conclude.

FILES

- 01-article-Overbaugh-2026.pdf ARTICLE

`cit-001257` · Scholar exhibit *reusable-glycan-microarrays-using-a-microwave-assisted-wet-e-citing-2026-10-30.pdf* (Independent Citation Record, not enclosed here)

Project not assigned

Glycan-mediated interactions of infectious hypodermal and hematopoietic necrosis virus (IHHNV) with host tissues in Pacific whiteleg shrimp (Penaeus vannamei)

Chanyarak Sombutkayasith, Atthaboon Watthammawut, Orawan Thongsum, Jeanne Delafontaine, Supawich Boonkua, Monsicha Somrit, Hiroto Hirayama, Somluk Asuvapongpatana, Clement Delannoy, Coralie Bompard
Glycobiology, 2026 · academic.oup.com

SECTION I — BACKGROUND

Cited Paper(s)	Differential recognition of oligomannose isomers by glycan-binding proteins involved in innate and adaptive immunity 2021
Additional citations from this citing paper	`cit-000540` — GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays (Materials and Methods, p.30) `cit-000987` — Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies (Materials and Methods, p.29) `cit-001259` — Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process (Materials and Methods, p.30)
Citing Paper	Glycan-mediated interactions of infectious hypodermal and hematopoietic necrosis virus (IHHNV) with host tissues in Pacific whiteleg shrimp (Penaeus vannamei) (article) Earliest public date: Not shown in PDF
Project Number	Not assigned — this row cites a paper other than the four tools, so no project applies. Confirm with the applicant.
Citing Authors' Institutions	Not recorded — confirm from the PDF's affiliation block.
Citing Article Type	article
Where the citation appears	**Discussion**, page 19 — Tier C — cited in a high-reputation publication

SECTION II — OVERVIEW OF USE

> *“receptors known to bind to oligomannosylated N-glycans exhibit, in fact, unique, and sometimes very restricted, specificities toward oligomannose motifs (Gao et al. 2021)”*

Discussion, p. 19 — Tier C — cited in a high-reputation publication

`cit-000540` — additional citation of GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays

> *“The results have analyzed with GLAD (Glycan Array Dashboard), which enables comparison of binding profiles and identification of enriched glycan motifs (Mehta and Cummings 2019)”*

Materials and Methods, p. 30 — Tier A — the tool is used in the Methods as the method performed

`cit-000987` — additional citation of Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies

> *“printed glycan microarray platform (version 5.6), comprising 539 glycans printed in six replicates per glycan (https://research.bidmc.org/ncfg/service-collaboration-request) (Heimburg-Molinaro et al. 2024)”*

Materials and Methods, p. 29 — Tier A — the tool is used in the Methods as the method performed

`cit-001259` — additional citation of Reusable glycan microarrays using a microwave assisted wet-erase (MAWE) process

> *“signal intensities were processed using the MAWE workflow, which excludes the highest and lowest replicate values before calculating the mean relative fluorescence units (RFU) standard deviation, and coefficient of variation for each glycan (Mehta et al. 2024)”*

Materials and Methods, p. 30 — Tier A — the tool is used in the Methods as the method performed

Objectives

Sombutkayasith, Watthammawut, Guérardel and colleagues set out to establish whether the infectious hypodermal and hematopoietic necrosis virus (IHHNV), which damages farmed Penaeus vannamei shrimp, attaches to host tissues through glycans. Their account of the work runs from the title and summary on p.1 through the results beginning on p.6, where they probe viral capsid and capsid-protein binding to glycan classes, and they conclude on p.3 that high-mannose N-glycans and fucosylated O-glycans act as primary attachment factors.

Importance to Field

Shrimp viral disease is an agricultural problem of real scale, and the paper's own framing on p.1 is that the entry route of IHHNV into host tissue was not understood. Establishing a glycan dependence gives the field a receptor-level hypothesis to test and, as the authors put it on p.3, points to glycan-mediated interaction as a target for intervention rather than leaving the virus's host range unexplained.

How and Why Your Work Was Used

The article's glycan microarray work is described under the heading "Glycan microarray" on p.29, where the authors state that binding specificity of IHHNV virus-like particles was evaluated on the National Center for Functional Glycomics printed glycan microarray platform (version 5.6) with 539 glycans in six replicates, citing the applicant's review of the platform and its glycan-interactome use (Heimburg-Molinaro, Mehta, Tilton and Cummings) as the source of the method. On p.30 they write that signal intensities "were processed using the MAWE workflow, which excludes the highest and lowest replicate values before calculating the mean relative fluorescence units (RFU) standard deviation, and coefficient of variation for each glycan (Mehta et al. 2024)" — that is, the applicant's published data-reduction rules for reusable arrays governed how every value in their figures was computed. The same paragraph on p.30 continues that the results were

"analyzed with GLAD (Glycan Array Dashboard), which enables comparison of binding profiles and identification of enriched glycan motifs (Mehta and Cummings 2019)", so GLAD was the tool through which the binding profiles were compared and the enriched motifs identified. Both instruments are named in Materials and Methods, in the paragraph that produces the paper's glycan-binding evidence, and the applicant's personal name appears in the in-text citation on both.

Findings / Conclusions / Recommendation

The authors report that the viral particles bind the glycan classes they expected for an attachment step, and on p.8 they state that the results "collectively strongly support the presence of complex glycans harboring terminal mannose, GlcNAc/GalNAc, fucose residues" on the host side. Their closing inference on p.3 is that high-mannose N-glycans and fucosylated O-glycans serve as primary attachment factors in IHHNV infection.

FILES

- 01-article-Guerardel-2026.pdf ARTICLE

`cit-000416` · additional rows cit-000540;cit-000987;cit-001259 · Scholar exhibit *differential-recognition-of-oligomannose-isomers-by-glycan-b-citing-2026-10-29.pdf* (Independent Citation Record, not enclosed here)

Appendix — candidates screened out

These citations were examined and are not argued in the packet. They are listed so the screening is on the record.

(unset) (0)

row_uid	Title	Senior author	Venue	Year

not independent (1)

row_uid	Title	Senior author	Venue	Year
cit-000511	Development of smart anti-glycan reagents using immunized lampreys	Cummings	Communications …	2020